

Lab Assignment 4

1. Purpose

Cryptanalysis of Shift Cipher using Brute Force, Dictionary Scoring and Chi-Square Analysis

2. Laboratory Notebook – What to Write

The laboratory notebook should contain the following sections:

1. Both the algorithms of cryptanalysis
2. Comparison: Compare the key predicted by dictionary scoring with the key predicted by Chi-Square analysis
3. Failure Analysis: If an attack fails, explain why it may have failed and suggest an improvement
4. Observations: Write observations based on your own experimental results
5. Conclusion: Summarize what was learned about the cryptanalysis of Shift Cipher

Students should maintain a results table similar to the following:

Test Case	Actual Key	Dictionary Key	Chi-Square Key	Dictionary Correct?	Chi-Square Correct?
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3. GitHub – Required Repository Structure

Continue using the existing CryptoLabX repository. The recommended structure is:

```
CryptoLabX/
├── attacks/
│   └── shift_cipher_attack/
│       ├── src/
│       │   ├── shift_cipher.py
│       │   ├── brute_force_dictionary.py
│       │   ├── chi_square_attack.py
│       │   └── main.py
│       ├── dictionary/
│       │   └── english_words.txt
│       ├── testcases/
│       ├── outputs/
│       ├── screenshots/
│       ├── reports/
│       │   └── Assignment_4_Report.pdf
│       └── README.md
```

4. Git Commit Requirement

You should maintain meaningful Git commit history rather than uploading the entire assignment as one final commit.

- Commit 1: Implemented Shift Cipher.
- Commit 2: Added brute-force and dictionary scoring.
- Commit 3: Added Chi-Square cryptanalysis.
- Additional commits may be used for testing, bug fixing and documentation.